Combinatorics II

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Abstract

Lecture note of Combinatorics II.

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Chapter 1

Sorting Algorithm

Lecture 1

24 Feb.

Problem 1.0.1. Given n distinct numbers $a_1, a_2, \ldots, a_n$, sort them in increasing order.

- Input: n distinct numbers $a_1, a_2, \ldots, a_n$.
- Output: a permutation $\pi \in S_n$ s.t.

$$a_{\pi(1)} < a_{\pi(2)} < \cdots < a_{\pi(n)}.$$

Observation. Given x, y, is $x < y$?

Remark 1.0.1. When analyzing an algorithm, we consider the worst-case input and count operations, where operations are defined according to context.

Algorithm 1.1: Brute force

Go through all $\pi \in S_n$ one by one and test each permutation to see if it is correct.

Remark 1.0.2. For running time, testing a permutation takes $\leq n-1$ comparisons, and there are $n!$ possible permutations, so

$$\# \text{ of comparisons} \leq n!(n-1).$$

Question. What is efficient? How does the running time scale as the input size n increases? Ideally, the running time should be polynomial $O(n^c)$ for some $c \in \mathbb{R}$.

Algorithm 1.2: Insertion Sort

Maintain a sorted partial list, add one element at a time until the entire list is sorted.

Algorithm 1.3: Binary insertion sort

```
// As before, we maintain a sorted prefix.
```

Insert a_i into sorted list $a_1 < a_2 < \cdots < a_{i-2} < a_{i-1}$. Compare a_i to the middle element. If similar, insert into the first half. If bigger, insert into the second half. `// Either way reduce number of possible positions by half each time`

Remark 1.0.3. Thus, we can insert in $\approx \log_2 i$ many comparisons each time, and thus in total we need $O(n \log_2 n)$ comparisons.

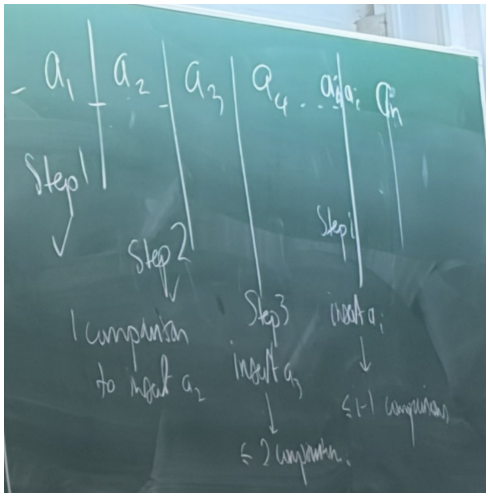
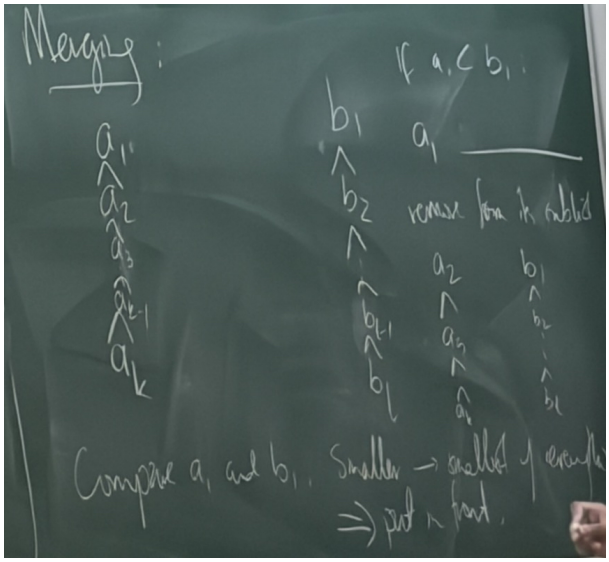


Figure 1.1: Insertion sort

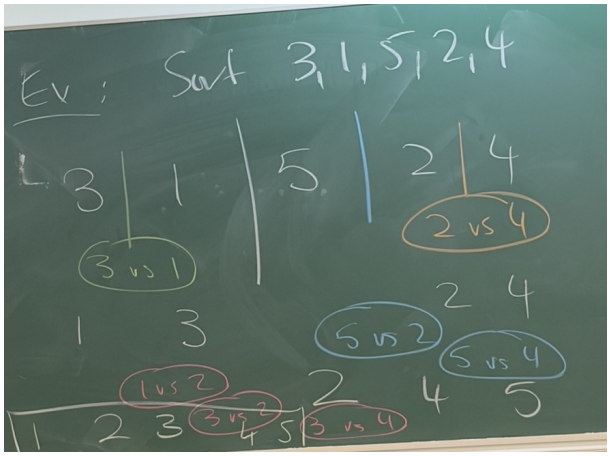
Algorithm 1.4: Merge Sort

```
// Recursive algorithm
```
Split the list into two equal halves.
Sort each half.
Merge the two sorted lists into one. `// Compare the head of two list, remove the smaller one from original list then put it in the back of a new list, which is empty initially, and repeat this comparison step.`

Remark 1.0.4. Number of comparison to merge $\leq n - 1$.



(a) Merging



(b) An example of merge sort

Remark 1.0.5. Running time: Let $T(n)$ be the worst case running time for merge sort on a list of

n numbers. Then $T(1) = 0$, and we know

$$T(n) \leq \underbrace{T\left(\lfloor \frac{n}{2} \rfloor\right)}_{\substack{\text{sorting the} \\ \text{first half}}} + \underbrace{T\left(\lceil \frac{n}{2} \rceil\right)}_{\substack{\text{sorting the} \\ \text{second half}}} + \underbrace{n-1}_{\text{merge}}.$$

Now suppose $n = 2^k$, $k \in \mathbb{N}$, then

$$\begin{cases} T(1) = 0 \\ T\left(2^k\right) = 2T\left(2^{k-1}\right) + 2^k - 1 \text{ for } k \geq 1. \end{cases}$$

Thus,

$$\begin{aligned} T\left(2^k\right) &= 2T\left(2^{k-1}\right) + 2^k - 1 \\ &= 2\left[2T\left(2^{k-2}\right) + 2^{k-1} - 1\right] + 2^k - 1 \\ &= \cdots = 2^i T\left(2^{k-i}\right) + i2^k - \left(2^i - 1\right), \end{aligned}$$

and if we let $i = k$, then

$$T\left(2^k\right) = 2^k T(1) + k2^k - 2^k + 1 = k2^k - 2^k + 1 = n \log_2 n - n + 1.$$

Thus, $T(n) \leq n \log_2 n - n + 1$ when $n = 2^k$.

Question (Lower bounds). Can we do better?

Answer. No, but to prove this, we need to show that any other algorithm, no matter how weird, takes as long. ⊛

Question. Suppose we have an algorithm A for sorting n numbers. Now can we lower-bound its runtime?

Information Theory approach

Each comparison gives ≤ 1 bit of information. At the end, we get the information of the permutation to sort the numbers, and since there are $n!$ possible permutations, so the number of time needed is $\geq \log_2 n!$.

Lecture 2

An arbitrary sorting algorithm is a function that takes the sequence of answers to our queries (is $x_i < x_j$?) and returns a permutation $\pi \in S_n$ s.t. $x_{\pi(1)} < x_{\pi(2)} < \cdots < x_{\pi(n)}$. Suppose the algorithm has a worst-case running time of k. Then this function has at most 2^k different outputs since such binary decision tree can has at most 2^k leaves and each leaf represent a possible permutation. 3 March.

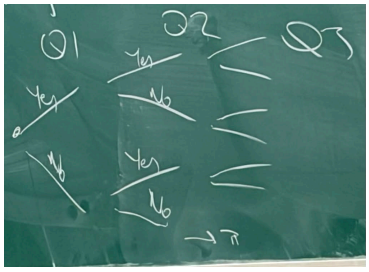


Figure 1.3: The decision tree

The entirety of S_n has to be in the image, so $2^k \geq |S_n| = n!$. Hence, $k \geq \log_2(n!) = (1 + o(1))n \log_2 n$.

Alternative arguments (playing against the devil)

You ask the comparisons to the devil, whose goal is to make the game go on as long as possible, The devil maintains a list of all possible permutations. Every time we do a comparison, we split the list into two: Those whose the answer is "yes" and those where it is "no". The devil chooses the answer with the bigger part. Game ends when the list has size 1. Let L_i be the list of possible permutations after i comparisons. $L_0 = S_n$, and for all $i \geq 1$,

$$|L_i| \geq \frac{1}{2}|L_{i-1}|.$$

Suppose one algorithm requires k comparisons in the worst case, then

$$1 \geq |L_k| \geq \frac{1}{2}|L_{k-1}| \geq \frac{1}{4}|L_{k-2}| \geq \cdots \geq \frac{1}{2^k}|L_0| = \frac{n!}{2^k},$$

so $2^k \geq n!$, which shows $k \geq (1 + o(1))n \log_2 n$.

Chapter 2

Connectivity and Spanning Tree

Definition 2.0.1 (graph). A (simple) graph is an ordered pair of sets (V, E), where V is a set of vertices and $E \subseteq \binom{V}{2}$ is a set of edges, where an edge is an unordered pair of disjoint vertices.

Definition 2.0.2. A walk in a graph is a sequence

$$v_0 e_1 v_1 e_2 v_2 e_3 v_3 \dots e_n v_n$$

where $v_i \in V$ and $e_i = \{v_{i-1}, v_i\} \in E$ for all i. A trail is a walk that does not repeat any edges. A path is a trail/walk without repeated vertices. A circuit is a trail that starts and ends at the same vertex. A cycle is a circuit that doesn't repeat any internal vertex.

Definition 2.0.3. A graph $G = (V, E)$ is connected if for every $u, v \in V$, there is a path(walk) from u to v.

Definition 2.0.4. Given a graph $G = (V, E)$ and a vertex $v \in V$, the connected componnent of v is the maximal connected subgraph of G that contains v.

Remark 2.0.1. $G' = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E \cap \binom{V'}{2}$. It is induced if $E' = E \cap \binom{V'}{2}$. It is spanning if $V' = V$.

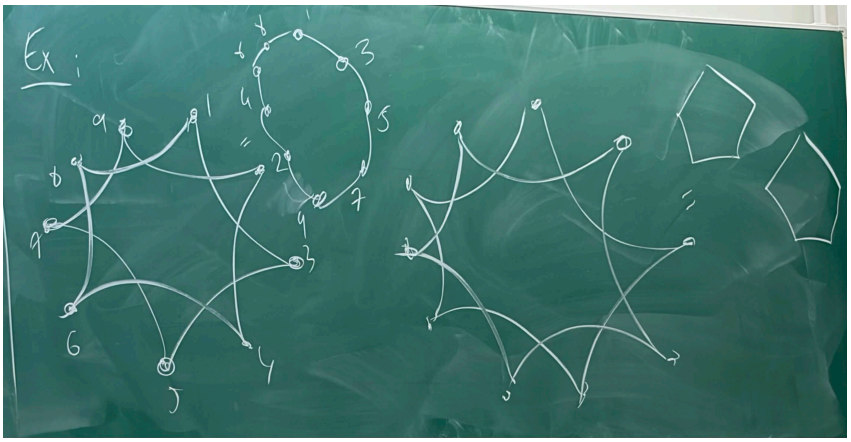


Figure 2.1: An example of connectivity (left one is connected but right one not)

Question. How can we identify the connected components of a graph?

Algorithm 2.1: Find the connected component containing v

Input: A graph $G = (V, E)$ and a vertex $v \in V$
Output: Connected component $C \subseteq V$ s.t. $v \in C$

```
// We will maintain two sets of vertices. Let W be the component so far, i.e. vertices we can reach from v whose neighborhood we have explored.
```

Initially $W_0 = \varnothing$. $Q_0 = \{v\}$. `// Q =queue of vertices to explore`
for *the i-th step* **do**
  if $Q_{i-1} = \varnothing$ **then**
    DONE. Return W as the component as v.
  else
    Let w be an arbitrary vertex in Q_{i-1}. $Q_i = Q_{i-1} \setminus \{w\}$, $W_i = W_{i-1} \cup \{w\}$.
    Explore the neighborhood of w: If u is a neighborhood of w and $u \notin Q \cup W_i$, then add it to Q_i.

Claim 2.0.1. The algorithm is finite.

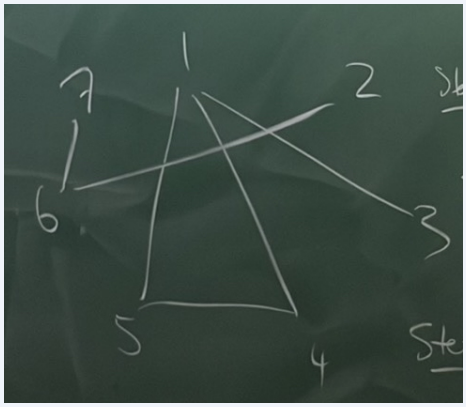
Proof. In every step, if we are not done, we add a distinct item to W (since Q_i and W_i are disjoint) and never remove it. Thus, we do at most $|V(G)|$ steps. ⊛

Claim 2.0.2. The algorithm is correct.

Proof. Let C_v be the component of v and W be the output of the algorithm. We show that $C_v = W$. We first show that $W \subseteq C_v$. Prove by induction on number of steps that we only add something to W if it was in Q. Also, we only add something to Q if it was a neighbor of an earlier member of W. By induction, there is a path from this vertex to v. Now we show $C_v \subseteq W$. Suppose not, then let $x \in C_v \setminus W$. Then there exists vx-path in G, say $v = v_0 v_1 v_2 \dots v_t = x$. Let j be the smallest index s.t. $v_j \notin W$. Then v_{j-1} is in W. Thus, we have explored its neighborhoods, so it would be in W. ⊛

It would be nice to not only know the components of v, but also paths that get there from v. Thus, in the algorithm, when exploring the neighborhood of a vertex $w \in Q_{i-1}$, if we add a neighbor u to Q_{i-1}, we keep track of the edge $\{w, u\}$ that was used to reach u.

Example 2.0.1. Now if $v = 4$ and the graph is the following figure:



Then we know

- Step 0: $Q_0 = \{4\}$, $W_0 = \{\}$.
- Step 1: $w = 4$, $Q_1 = \{1(\{1, 4\}), 5(\{4, 5\})\}$, and $W_1 = \{4\}$.
- Step 2: $w = 1, Q_2 = \{5(\{4, 5\}), 3(\{1, 3\})\}$, and $W_2 = \{4, 1(\{1, 4\})\}$.

- Step 3: $w = 5, Q_3 = \{3(\{1,3\})\}$, and $W_3 = \{4, 1(\{1,4\}), 5(\{4,5\})\}$.
- Step 4: $w = 3, Q_4 = \{\}$, and $W_4 = \{4, 1(\{1,4\}), 5(\{4,5\}), 3(\{1,3\})\}$.

Definition 2.0.5. A tree is a minimally connected graph, i.e. it is connected, but removing any edge disconnect it.

Proposition 2.0.1. Let T be an n-vertex graph. The following is equivalent:

(1) T is a tree.

(2) T is maximally acyclic.

(3) T is connected and has $n-1$ edges.

(4) T is acyclic and has $n-1$ edges.

Remark 2.0.2. Acyclic means no cycle.

Lecture 3

If we modify Algorithm 2.1 so that we take the shortest element in Q, then we call this algorithm **Breadth-first search**. 6 March.

If we modify Algorithm 2.1 so that we take the newest element in Q, then we call this algorithm **Depth-first search**.

2.1 Minimum weight spanning tree

Theorem 2.1.1 (Cayley)**.** The complete graph K_n has n^{n-2} spanning trees.

We want to find the most efficient/cheapest spanning tree.

Problem 2.1.1 (Mininimum spanning tree)**.**

- Input: Complete graph K_n and a weight function $w : E(K_n) \to \mathbb{R}_{\geq 0}$
- Output: A spanning tree T of K_n minimizing $\sum_{e \in E(T)} w(e)$.

Example 2.1.1. If G is an n-vertex graph (not necessarily complete). Define

$$w(e) = \begin{cases} 0, & \text{if } e \in E(G); \\ 1, & \text{if } e \notin E(G). \end{cases}$$

Then G is connected if and only if there exists a tree of weight 0.

Algorithm 2.2: Kruskal's algorithm(Theoretical way)

```
// Maintain F, and let F = ∅ initially.
Order the edges by weight, say e_1, e_2, ..., e_m.
for each i s.t. 1 ≤ i ≤ m do
    if F + e_i contributes a cycle then
        do nothing.
    else
        add e_i to F.
```

Algorithm 2.3: Kruskal's algorithm(Implementation way)

```
// Maintain F, and let F = ∅ initially, and C_v is the component containing v
   for all v ∈ V(G), where C_v = {v} initially for all v ∈ V(G).
Order the edges by weight, say e_1, e_2, ..., e_m.
for each i s.t. 1 ≤ i ≤ m do
    Let e_i = (u, v). if C_u = C_v then
        do nothing.
    else
        add e_i to F.
        Merge C_u and C_v.
```

Claim 2.1.1. At the end of the Kruskal's algorithm, F is a minimum spanning tree.

Proof. We have to show F is a spanning tree and F is minimal.

We first show F is a spanning tree. It suffices to show F has no cycles and connected. Note that in this algorithm we make sure every adds of an edge into F would not contribute a cycle, so the final F must has no cycle. Now we show F is connected. If not, then we have connected components. Since G is connected, there will be edges between components. Thus the algorithm would have added the first edge between these components. Now we know F is connected and acyclic. If F does not cover all vertices, say not covering v, then the edge between v and F would be added in the algorithm, so F covers all n vertcies, i.e. F is a spanning tree.

Now let T_k be the tree from Kruskal's algorithm. Let $T_{\min}$ be a minimum spanning tree that maximizes $|E(T_k) \cap E(T_{\min})|$. If $|E(T_k) \cap E(T_{\min})| = n - 1$, then T_k is a minimum spanning tree. Otherwise, consider the first edge (i.e. e_v with minimal v) in $T_k \Delta T_{\min}$, say e_i.

Claim 2.1.2. $e_i \in T_k$.

Proof. If $e_i \in T_{\min}$, then in the i-th step of Kruskal algorithm, we would have added e_i to T_k. (For $e_1, e_2, \ldots, e_{i-1}$, each edge is either in both tree or not in any of both, so e_i can be added to T_k in the algorithm since it will not contribute to a cycle (since $T_{\min}$ has no cycle)). Thus, $e_i \in T_k$. ⊛

Consider $T_{\min} + e_i$, which creates a cycle C. Thus, C must contain an edge e_j with $j > i$ (because all edges e_j with $j < i$ are common to both trees, and $C \nsubseteq T_k$). Let $T' = T_{\min} + e_i - e_j$, then T' is a spanning tree with smaller weight sum, which is a contradiction. ⊛

Lecture 4

Running time of Kruskal's algorithm

10 March.

First, sorting all edges by weight takes $O(m \log_2 m)$ steps. Then, iterate through all edges, if $c_u = c_v$, do nothing, and this steps take $O(1)$ steps. If not, then add edges and update components, which takes $O(n)$ steps, and will run this case $n - 1$ times. Thus, the total time:

$$O(m \log_2 m) + O(m) + O\left(n^2\right) = O(m \log_2 m + n^2) = \begin{cases} O(m \log m), & \text{if } m = \Omega\left(\dfrac{n^2}{\log n}\right); \\ O\left(n^2\right), & \text{if } O\left(\dfrac{n^2}{\log n}\right). \end{cases}$$

With more improvement,

$$O(m \log m) \text{ for all } m.$$

Chapter 3

Dijkstra's Algorithm

Drawbacks of trees as networks

(1) Unique path between any two vertices, which could be unnecessarily long.

(2) No redundancy/robustness: removing any one edge/vertex disconnects the network.

Question. Given a network $G = (V, E)$ and weights $w : E \to \mathbb{R}_{\geq 0} \cup \{\infty\}$, and given two vertices $u, v \in V$, how do we find the cheapest path from u to v?

Observation. We may assume G is complete. If an edge $\{x, y\}$ is missing, add it with ∞ weight.

Lemma 3.0.1. If P is a cheapest path from u to v, then for any $x, y \in P$, $P[x, y]$ is a shortest path from x to y.

Algorithm 3.1: Dijkstra's algorithm

Data: A complete graph $G = (V, E)$.
Result: For any vertex x, the cheapest graph from u to x.
Idea: Keep a set W of vertices which we have already found the cheapest path from u. For other vertices, keep track of a tentative cheapest path, i.e. the cheapest path we have found so far.

```
W ← {u}, t(u) = 0, prev(u) = ∅.
For any x ∈ V \ {u}, t(x) = w({u, x}), prev(x) = u.
while W ≠ V do
    Let y = arg min_{x∈V\W} t(x), i.e. t(y) minimizes t(x) for all x ∉ W.
    Add y to W.
    for every x ∉ W ∪ {y} do
        if t(x) > t(y) + w({y, x}) then
            t(x) ← t(y) + w({y, x}), prev(x) = y.
Cheapest path from x to u is x, prev(x), prev(prev(x)), ..., u with cost t(x).
```

Theorem 3.0.1. Dijkstra's algorithm terminates after $n - 1$ rounds and returns shortest path.

Proof. Each round adds a new vertex to W, which never gets removed, so after $n - 1$ rounds, $W = V$. For correctness, we can do induction on the number of rounds:

Claim 3.0.1. At the beginning of the i-th iterartion:

(a) $|W| = i$.

(b) for every $x \in V$, the vertices $x, \text{prev}(x), \text{prev}(\text{prev}(x)), \dots, u$ form a path from x to u of weight $t(x)$, where all vertices except x lie in W.

(c) for any vertex $z \in V \setminus W$ with $t(z) = \min\{t(x) : x \in V \setminus W\}$, $t(z)$ is the minimum cost of a uz-path.

(d) For every $y \in V \setminus W$ and $v \in W$,

$$t(y) \leq t(v) + w(\{v, y\}).$$

Proof.

- Base case: $i = 1$, (a), (b), (d) are obvious. For (c), suppose not, then if z minimize $t(\cdot)$ but there is a cheaper uz-path, say

$$u \to x \to \cdots \to z \in P,$$

then

$$w(P) = w(\{u, x\}) + w(\{x, \dots\}) + \cdots \geq w(\{u, x\}) = t(x) \geq t(z),$$

which is a contradiction.

- Induction step: (a) is obvious.

Now we show (b). If we didn't update $t(x), \text{prev}(x)$ in the $(i-1)$-th iteration, then this is true by induction. If we did update, then $\text{prev}(x) = v_0$ where v_0 is the vertex added to W in the $(i-1)$-th iteration, and $t(x) = t(v_0) + w(\{v_0, x\}) =$ weight of path $u \to \cdots \to v_0 \to x$, and u to $\text{prev}(v_0)$ all in W by induction, and v_0 was added to W in the $(i-1)$-th iteration.

Now we show (c). Suppose not, then $\exists z \in V \setminus W$ minimizing $t(z)$ but $t(z) > d(z, u)$. Consider the cheapest path from u to z, say P. Let y be the first vertex in P (starting from u) that is not in W. (such y exists since z is not in W) Let x be the preceding vertex on P, noting that $x \in W$. Then,

$$t(z) > w(P) = w(P_{[u,x]}) + w(\{x, y\}) + w(P_{[y,z]}) \geq w(P_{[u,x]}) + w(\{x, y\}).$$

By Lemma 3.0.1, $P_{[u,x]}$ has to be a lightest path from u to x. By induction (where we add x to W), $t(x) = w(P_{[u,x]})$. Thus,

$$t(z) > w(P_{[u,x]}) + w(\{x, y\}) = t(x) + w(\{x, y\}) \geq t(y),$$

which is by the induction statement of (d). However, this contradicts z minimizeing $t(v)$ for $v \in V \setminus W$.

As for (d), after the step where we added v to W, we check whether $t(v) + w(\{v, y\}) < t(y)$, and update if so. Hence, after this round, $t(v)$ will not change, and $t(v)$ is decreasing, so it is impossible that $t(y) > t(v) + w(\{v, y\})$.

⊛

This implies the correctness: for any x, at the iteration when we add x to W, (c) implies we have the correct path for x and we never change it subsequently. ■

Lecture 5

13 March.

3.1 Running Time of Dijkstra's Algorithm

Since we have $n-1$ iterations, and in each iteration, we have to find the minimum, which takes $O(n)$ steps, and updating also takes $O(n)$ steps, so the total running time is $O\left(n^2\right)$.

3.2 Traveling Salesman Problem

Problem 3.2.1 (Traveling Salesman Problem)**.** If we have a weighted graph $G = (V, E)$, where each vertex is a city and each edge is a road between two cities, and we are at $v \in V$ initially. Then, how to travel to every city exactly once and then return to the starting point v while minimizin the cost?

Definition 3.2.1 (Hamilton cycle)**.** A cycle that goes through every vertex in a graph is called a Hamilton cycle.

Remark 3.2.1. Traveling Salesman Problem's goal is to find a Hamilton cycle of minimum cost of the given graph.

Observation. The number of possible routes is $\frac{1}{2}(n-1)!$ since we can just permutate the $n-1$ vertices that is not the starting point, and note that the direction is not important, so $u \to a \to b \to u$ and $u \to b \to a \to u$ is considered same routes.

Thus, the brute force takes $O(n) \cdot \frac{1}{2}(n-1)! = O(n!)$ time since for each route we have to go through this path to compute the cost.

Remark 3.2.2. The Held-Karp Algorithm, which uses dynamic programming, takes $O\left(n^2 2^n\right)$ time to solve Traveling Salesman Problem (also exponential in memory).

Question (Open question)**.**

- Is there a polynomial time algorithm?
- Is there an algorithm takes $O(1.99999999^n)$ time?

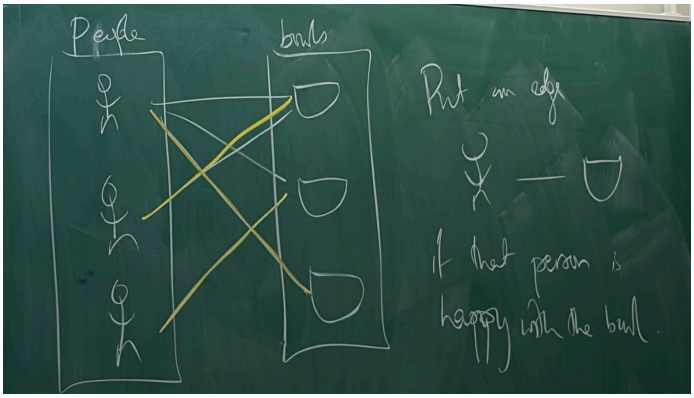
Chapter 4

Matchings

Lecture 6

Example 4.0.1. n friends going for dinner. n bowls of beef noodle soup served. Each person has a subset of the bowls they would be happy with. Can we give one bowl to each person and making every one happy? 17 March.

Proof. We can build a graph as follow:



⊛

Definition 4.0.1. A graph G is bipartite if we can partition $V(G) = X \cup Y$ s.t. for every $e \in E(G)$,

$$|e \cap X| = 1 = |e \cap Y|.$$

Definition 4.0.2. A matching in a graph is a set of pairwise-disjoint edges. If a vertex is in an edge of the matching, it is called "saturated". If every vertex is saturated, the matching is perfect.

Observation. If there is some $S \subseteq X$ with $|N(S)| < |S|$, then there is no matching saturating X.

Proof. If there is a matching saturating X and $\exists S \subseteq X$ with $|N(S)| < |S|$, then each vertex in S would match with some vertex in $N(S)$ in the matching saturating X. Note that every vertex in S cannot match to same vertex. Hence, $|N(S)| \geq |S|$, which is a contradiction. ■

Theorem 4.0.1 (Hall's Marriage Theorem)**.** Let $G = (X \cup Y, E)$ be a bipartite graph. If $|N(S)| \geq |S|$ for all $S \subseteq X$, then G has a matching saturating X.

Proof. We can do induction on $|X|$.

- Base case: $|X| = 1$. Let $S = X$, and $|N(X)| \geq |X| = 1$, then there exists $y \in Y$ s.t. $\{x, y\} \in E$, so we can pick this edge to form a matching saturating X.
- Induction Step: $|X| \geq 2$.
  - Case 1: $\forall \varnothing \neq S \subsetneq X$, $|N(S)| > |S|$. Let $\{x, y\}$ be an arbitrary edge, where $x \in X$ and $y \in Y$. Let $G' = (X' \cup Y', E')$ where $X' = X - \{x\}$, $Y' = Y - \{y\}$, and $E' = E|_{X' \times Y'}$. We use induction to find a matching M' in G' saturating X'. Then, $M = M' \cup \{x, y\}$ is a matching saturating X. Note that we can use induction hypothesis since for $S \subseteq X'$, $N_{G'}(S) = N_G(S) \setminus \{y\}$. Hence,
    $$|N_{G'}(S)| \geq |N_G(S)| - 1 \geq |S|,$$
    and thus we can find M'.
  - Case 2: there exists some $\varnothing \neq X_1 \subsetneq X$ with $|N(X_1)| = |X_1|$. Let $N(X_1) = Y_1$. Let $X_2 = X \setminus X_1$, and $Y_2 = Y \setminus Y_1$, and $G_1 = G[X_1 \cup Y_1]$. By induction, G_1 has a matching M_1 saturating X_1 since for all $S \subseteq X_1$, we have $S \subseteq X$ and thus
    $$|N_G(S)| \geq |S|,$$
    and note that $N_{G_1}(S) = N_G(S)$ since $N_G(S) \subseteq N_G(X_1) = Y_1$. If we can find a matching M_2 in $G_2 = G[X_2 \cup Y_2]$, then $M = M_1 \cup M_2$ is a matching saturating X. Let $S \subseteq X_2$. Consider $S' = S \cup X_1$. By assumption, $|N_G(S')| \geq |S'|$. Note that $|S'| = |S| + |X_1|$. Hence,
    $$N_G(S') = N_G(S \cup X_1) = Y_1 \cup \underbrace{N_{G_2}(S)}_{\subseteq Y_2}.$$
    Note that $N_G(S') = Y_1 \cup N_{G_2}(S)$ is a disjoint union since for $N_G(S)$, it is either in Y_1 or not, if not, then it is in $N_{G_2}(S)$. This shows
    $$|S| + |X_1| = |S'| \leq |N_G(S')| = |Y_1| + |N_{G_2}(S)| = |X_1| + |N_{G_2}(S)|.$$
    Hence, $|N_{G_2}(S)| \geq |S|$ for all $S \subseteq X_2$. By induction, we can find a matching M_2 in G_2 saturating X_2.

■

Corollary 4.0.1. If G is a d-regular bipartite graph, then G has a perfect matching.

Proof. Given any $S \subseteq X$,
$$d|S| = \#\{\text{edges between } S, N(S)\} \leq d|N(S)|$$
since for vertices in $N(S)$, they may be adjacent to vertex not in S, so by Hall's theorem, there is a matching saturating X, which is a perfect matching since
$$d|X| = \#\{\text{edges between } X, Y\} = d|Y|,$$
and thus $|X| = |Y|$. ■

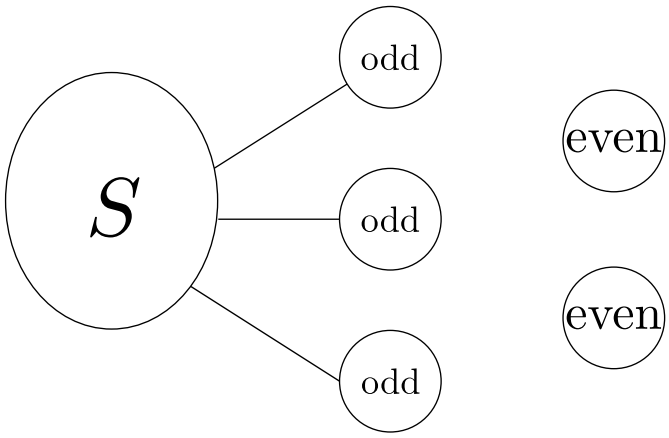
Question. What are necessary and sufficient conditions to have a perfect matching in general (i.e. non-bipartite) graph?

For necessary condition:

- n should be even.
- every vertex should has positive degree.

- every connected components should be even.
- for any $S \subseteq V(G)$, $o(G-S) \leq |S|$, where $o(G-S)$ is the number of odd components in $G-S$.

Remark 4.0.1. For the last argument, it is because G has a perfect matching, and for the odd components in $G-S$, there must be edges between them and S, i.e. some vertices in those odd componenets must match with the vertex in S, and thus $o(G-S) \leq |S|$.



Theorem 4.0.2 (Tutte, 1947)**.** If $o(G-S) \leq |S|$ for every $S \subseteq V(G)$, then G has a perfect matching.

proof (Lovasz, 1975). Note that for $S = \varnothing$, we have $o(G) \leq |0|$, so $|V(G)|$ is even. If G is complete, then G obviously has a perfect matching. Now we consider the case that G is not complete. Suppose for contradiction. G is a counterexample on n vertices with the maximum number of edges.

Claim 4.0.1. $\forall \{x, y\} \notin E(G)$, $G' = G + \{x, y\}$ has a perfect matching.

Proof. We have to show G' satisfies Tutte's condition. If so, then since G was a longest counterexample, G' cannot be a counterexample, so G' has a perfect matching.

Note that we have $o(G-S) \leq |S|$ for all $S \subseteq V(G)$. If $\{x, y\} \cap S \neq \varnothing$, then $o(G'-S) = o(G-S)$. If $\{x, y\}$ is contained in a component of $G-S$, components don't change, so $o(G'-S) \leq |S|$. If $\{x, y\}$ connects two components of $G-S$, then no matter what happens,

$$o(G'-S) \leq o(G-S) \leq |S|.$$

⊛

Remark 4.0.2. The perfect matching must pick $\{x, y\}$, otherwise we have a perfect matching in G.

Let $U = \{v \in G : \deg(v) = n-1\}$.

- Case 1: Every component of $G-U$ is a clique(complete subgraph). Each even clique admits a perfect matching. Each odd clique has a matching with only one unsaturated vertex. By Tutte's condition, number of odd cliques $\leq |U|$, say there are k odd cliques. Thus, we can match the unsaturated vertices to different vertices in U. Note that there will be even number

of unused vertices in U since

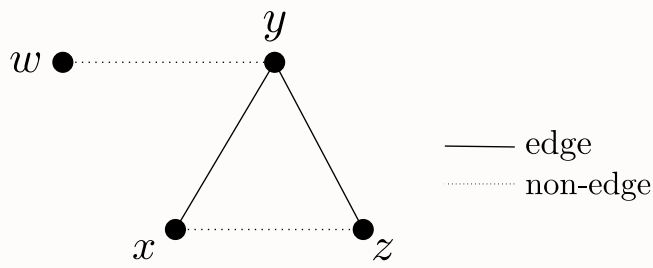
$$|V(G)| = |U| + \sum \text{size of components},$$

and since $|V(G)|$ is even, we have

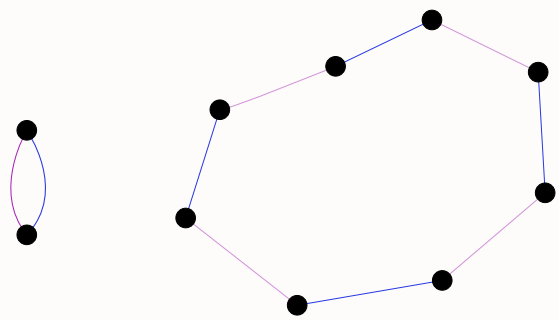
$$|U| \equiv k \mod 2.$$

This shows $|U| - k$ is even, i.e. after we match the unsaturated vertex in each odd component to U, the number of unused verteices in U is even, and thus we can match them arbitrarily.

- Case 2: There is a non-clique component in $G \setminus U$. We can find the following configuration:



The reason is in $G \setminus U$, there must be x, y, z which do not form a triangle, otherwise every component is a clique. On the other hand, $y \notin U$, so there exists some w that is not adjacent to y. Let M_1 be a perfect matching in $G + \{w, y\}$ and let M_2 be a perfect matching in $G + \{x, z\}$. Let $G' = M_1 \cup M_2$. Then the components of G' are either isolated edges (really in G) or even edges, alternating M_1 and M_2 edges.



To build a matching M of G, we can take all edges that are components in G'. Consider the cycle component C in G'. If the edges $\{y, w\}$ is not in C, then take the M_1 edges. If the edge $\{x, z\}$ is not in C, then take the M_2 edges. If both edges are in C: If we delete y from this cycle, then there remains a path with an even number of edges and odd number of vertices. We can delete one of x or z s.t. the two remaining paths both have an even number of vertices. Note that we have $\{y, z\}, \{y, x\}$ these two edges, so we can take the edges incident to the deleted pair and since the remaining two subpaths are of even vertices, so we can take alternating edges on two subpaths to form a matching.

■

Lecture 7

20 March.

Theorem 4.0.3 (Dirac, 1952)**.** Let G be an n-vertex graph with minimum degree at least $\frac{n}{2}$, then G has a Hamilton cycle.

proof of Dirac's theorem. We first introduce a lemma.

Lemma 4.0.1. Let G be an n-vertex graph with minimum degree $\geq \frac{n}{2}$. Let P be a longest path in G, then

(1) If u, v are the endpoint of P, then $N(u) \cup N(v) \subseteq V(P)$.

(2) There is a cycle C with $V(C) = V(P)$.

Proof.

(1) If an endpoint has a neighbor not in P, we can extend P by using the edge incident to that endpoint and that neighbor, which is a contradiction.

(2) If $\{u, v\} \in E$, add it to the path to get a cycle. Otherwise, let

$$P = u = v_1 \to v_2 \to v_3 \to v_4 \to \cdots \to v_i \to v_{i+1} \to \cdots \to v_m = v.$$

If $\exists i$ s.t. $\{v_i, v\}, \{u, v_{i+1}\} \in E$, then take C to be

$$u \to v_{i+1} \to v_{i+2} \to \cdots \to v_{m-1} \to v \to v_i \to v_{i-1} \to \cdots \to v_2 \to u.$$

Such an i must exist: let $S = \{v_i : \{u, v_{i+1}\} \in E\}$, then $|S| = \deg(u) \geq \frac{n}{2}$. Also, $|N(v)| = \deg(v) \geq \frac{n}{2}$. Note that

$$|S \cup N(v)| \leq |\{u, v_2, \ldots, v_{m-1}\}| \leq n - 1.$$

Thus, $S \cap N(v) \neq \varnothing$.

■

If we have this lemma, then let P be a longest path and v is an endpoint. By (1), $N(v) \subseteq N(P)$. Hence, $|V(P)| \geq \frac{n}{2} + 1$(contain all neighbors of V and v itself). By (2), there is a cycle C with $V(C) = V(P)$. If $V(P) = V(G)$, then C is a Hamilton cycle. Otherwise, let $x \in V(G) \setminus V(C)$, since $|V(C)| = |V(P)| > \frac{n}{2}$, and $N(x) \geq \frac{n}{2}$ and $|V(C) \cup N(x)| \leq n$, so $N(x) \cap V(C) \neq \varnothing$, i.e. there is a neighborhood of x in C. Then we find a path longer than P, contradicting our choice of P. ■

Remark 4.0.3. This minimum degree condition is sufficient but not necessary, e.g. $G = C_n, n \geq 5$.

Remark 4.0.4. The minimum degree condition is best possible. Otherwise, consider a bipartite graph $G = (U \cup V, E)$, where $|U| = \frac{n}{2} - 1$ and $|V| = \frac{n}{2} + 1$, and for each $u \in U$ and $v \in V$ we have $\{u, v\} \in E$. Thus, this subgraph can not have a Hamilton cycle.

Theorem 4.0.4 (Ore). Let G be an n-vertex graph, such that whenever $\{u, v\} \notin E(G)$, we have $\deg(u) + \deg(v) \geq n$, then G has a Hamilton cycle.

Appendix